

Smart and Sustainable Cars: Racing into the Future

Cars are getting smarter, is an understatement. Cars are getting über-smart, is the truth. All aspects of the modern car, from the powertrain to infotainment, are wedded to electronics, communications technology and a good deal of IT too, in the journey towards making driving—even on Indian roads—a really safe, pleasurable and sustainable experience. Here, in this article, we look at some of the multi-faceted roles a modern car plays



Mahindra Reva, solar charging

JANANI GOPALAKRISHNAN VIKRAM

My car, my guide

One look at a modern car's dashboard is enough to show the amount of vehicle management and driving assistance features are there in it! Advanced driver assistance systems (ADAS) are beginning to feature in many new-gen cars. A blanket term that covers everything from GPS-based navigation aids and traffic updates to adaptive cruise control (ACC), lane departure warning, blind spot detection, collision

avoidance, speed adaptation and night vision, ADAS basically covers a growing number of tools that improve situational awareness and understand the driver's temperament and automate routine tasks to make driving more convenient and safe.

"Automakers are trying hard to match consumer experience in cars to what they get using smartphones and tablets. This has resulted in high-end infotainment technologies getting integrated into today's cars. The focus on driver and passenger safety is driv-

ing developments in ADAS, based on radar and vision, super data fusion technologies to name a few," says Sanjay Gupta, director—Automotive MCU Group, India Design Centre, Freescale Semiconductors India Pvt Ltd. He cites some interesting examples, such as Volvo's City Safety, a low-speed crash avoidance system available with its XC60 crossover SUV. The detection system uses LIDAR, a cross between laser and radar, to prevent fender-benders below 14.5 kmph.

ACC is also emerging strong. Toyota, Ford, Mercedes and Jaguar already offer this facility on some vehicles. It tracks the car ahead, slowing down and speeding up automatically to maintain a safe distance in variable highway traffic.

"Our 77GHz radar systems support ACC, pre-crash protection and collision warning systems with and without automatic steering and braking intervention. In a collision warning system, the radar chipset can detect and track objects, automatically adjusting the vehicle's speed and distance in response to the traffic ahead and triggering a driver warning of an imminent collision and initiate emergency braking intervention," explains Gupta.

Although still expensive, today we also have technologies that sense when the driver is too tired to drive or has had too much to drink. Volvo, for instance, uses available lane departure sensing technology to detect when sleepy drivers make jerky, telltale steering wheel corrections. The system